Project Report

1. Introduction

This report presents a data quality audit and behavioral analysis of recent customer sign-up data for Rapid Scale, a SaaS company offering tiered subscription plans.
The objective was to:

- Assess data accuracy and completeness.
- Identify user acquisition patterns.
- Analyze subscription preferences
- Review marketing opt-in behavior and demographics.

The dataset initially contained 298 customer records. Each record included acquisition source, region, subscription plan, marketing status, age, and gender.
Before conducting trend analysis, a structured data cleaning process was completed to ensure reliable outputs.

2. Data Cleaning Summary

Several data quality issues were identified and addressed prior to analysis.

Invalid Dates:

- Some records contained invalid values such as “not a date” in the signup_date column. These values prevented correct datetime conversion.
- Six records with invalid dates were removed to ensure accurate time-based analysis.
-

[17]:
```
df_raw[df["signup_date"].isnull()][["customer_id", "signup_date"]]
```

[17]:

	customer_id	signup_date
0	CUST00000	NaN
80	CUST00080	NaN
120	CUST00120	not a date
159	CUST00159	not a date
197	CUST00197	not a date
217	CUST00217	not a date

Missing Primary Identifiers:

Two records contained missing customer_id values. As customer_id functions as the primary identifier, these records were removed to prevent duplication and reporting errors.

```
[22]: df[df.duplicated(subset="customer_id", keep=False)]
```

[22]:

	customer_id	name	email	signup_date	source	region	plan_selected	marketing_
161	NaN	Robert Carter	robert61@example.com	2024-06-10	LinkedIn	South	Pro	
287	NaN	Antonio Hammond	antonio87@inboxmail.net	2024-10-14	Instagram	West	prem	

Inconsistent Plan Values:

The plan_selected column contained inconsistent entries such as:

- basic
- PRO
- PREMIUM
- UnknownPlan

These were standardized to:

- Basic
- Pro
- Premium

Invalid entries such as “UnknownPlan” were treated as missing and removed to reserve analytical integrity.

```
[27]: df["plan_selected"].unique()
```

```
[27]: array(['basic', 'PREMIUM', 'Pro', 'Premium', 'UnknownPlan', 'PRO',
             'Basic', nan], dtype=object)
```

```
[28]: df["plan_selected"] = df["plan_selected"].str.strip().str.title()
```

```
[29]: df["plan_selected"].unique()
```

```
[29]: array(['Basic', 'Premium', 'Pro', 'Unknownplan', nan], dtype=object)
```

```
[30]: df["plan_selected"] = df["plan_selected"].replace("Unknownplan", np.nan)
```

```
[31]: df["plan_selected"].unique()
```

```
[31]: array(['Basic', 'Premium', 'Pro', nan], dtype=object)
```

```
[32]: df["plan_selected"].isnull().sum()
```

```
[32]: np.int64(14)
```

```
[33]: df.shape
```

```
[33]: (292, 10)
```

```
[34]: df = df.dropna(subset=["plan_selected"])
```

```
[35]: df["plan_selected"].isnull().sum()
```

Age Data Issues:

The age section had text values like "unknown" and "thirty", as well as an absurd number of 206.
The column was changed to a numeric format. Invalid entries were classified as missing. Unrealistic values were deleted. Missing data were replaced with the median age to ensure distribution stability.
After correction, the age range was 21 to 60, with a median of 34.

Gender and Marketing Inconsistencies:

Gender values had inconsistent casing, such as "male" and "FEMALE". These were standardized. An invalid entry ("123") was marked as missing.
The marketing_opt_in column contains an incorrect value ("Nil") that was changed to missing.

Region Data:

There were missing values in the region data. Rather than eliminating these records, missing entries were labelled as "Unknown" to save useable data while explicitly showing partial information.

[37]:
```
print("Missing region:", df["region"].isnull().sum())
print("Missing email:", df["email"].isnull().sum())
print("Missing age:", df["age"].isnull().sum())
print("Missing gender:", df["gender"].isnull().sum())
print("Missing marketing:", df["marketing_opt_in"].isnull().sum())

print("\nUnique gender values:", df["gender"].unique())
print("\nUnique marketing values:", df["marketing_opt_in"].unique())
print("\nUnique source values:", df["source"].unique())
```

```
Missing region: 28
Missing email: 31
Missing age: 12
Missing gender: 7
Missing marketing: 9

Unique gender values: ['Male' 'Non-Binary' 'Other' 'male' 'FEMALE' 'Female' nan '123']

Unique marketing values: ['Yes' 'No' nan 'Nil']

Unique source values: ['LinkedIn' 'Google' 'YouTube' 'Facebook' 'Referral' 'Instagram' nan '??']
```

[38]:
```
df["gender"] = df["gender"].str.strip().str.title()
```

[39]:
```
df["gender"] = df["gender"].replace("123", np.nan)
```

[40]:
```
df["gender"].unique()
df["gender"].isnull().sum()
```

[40]:
```
np.int64(13)
```

```
[41]: df["marketing_opt_in"] = df["marketing_opt_in"].replace("Nil", np.nan)

[42]: df["marketing_opt_in"].isnull().sum()

[42]: np.int64(10)

[43]: df["source"] = df["source"].replace("??", np.nan)

[44]: df["source"].isnull().sum()

[44]: np.int64(13)

[45]: df["region"] = df["region"].fillna("Unknown")

[46]: df["age"] = pd.to_numeric(df["age"], errors="coerce")

[47]: df.loc[df["age"] > 100, "age"] = np.nan

[48]: df["age"] = df["age"].fillna(df["age"].median())

[49]: df["age"].isnull().sum()
      df["age"].describe()

[49]: count    278.000000
      mean      35.410072
      std       10.679994
      min       21.000000
```

Final Dataset:

Original records: 298
Final records: 278
20 records removed due to critical data quality issues
The dataset is now structurally consistent and suitable for business analysis.

3. Key Findings & Trends

Subscription Plan Distribution:

The Premium plan has the highest number of sign-ups (97), followed closely by the Pro plan (93) and Basic plan (88).

Although the difference between tiers is relatively small, this suggests that many users are willing to adopt mid-to-high subscription tiers rather than only entry-level pricing.

Acquisition Sources:

Digital acquisition channels generate most new users. YouTube provides the highest number of sign-ups (53), followed by Google (48) and referral sources (46). This indicates that video-based marketing and search visibility play an important role in customer acquisition.

Demographic Profile:

The median customer age is 34.
Most users fall within the 25–40 age range.
This suggests the platform appeals primarily to early and mid-career professionals.

[59]:
```python
#Sign-ups by Source
df["source"].value_counts()
```

[59]:
```
source
YouTube      53
Google       48
Referral     46
Instagram    42
Facebook     39
LinkedIn     37
Name: count, dtype: int64
```

[60]:
```python
#Sign-ups by Region
df["region"].value_counts()
```

[60]:
```
region
East       58
North      57
South      55
West       42
Central    38
Unknown    28
Name: count, dtype: int64
```

[61]:
```python
#Sign-ups by Plan
df["plan_selected"].value_counts()
```

[61]:
```
plan_selected
Premium    97
Pro        93
Basic      88
Name: count, dtype: int64
```

[62]:
```
#Marketing Opt-In Counts by Gender
pd.crosstab(df["gender"], df["marketing_opt_in"])
```

[62]:

marketing_opt_in	No	Yes
gender		
Female	41	39
Male	49	33
Non-Binary	20	17
Other	32	24

[63]:
```
#Age Summary
print("Min:", df["age"].min())
print("Max:", df["age"].max())
print("Mean:", df["age"].mean())
print("Median:", df["age"].median())
print("Null count:", df["age"].isnull().sum())
```

```
Min: 21.0
Max: 60.0
Mean: 35.410071942446045
Median: 34.0
Null count: 0
```

4. Business Question Answers

1. Which acquisition source brought in the most users last month?

- Google generated the highest number of sign-ups in the most recent month analyzed. This indicates strong search visibility and/or campaign effectiveness within this channel.

2. Which region shows signs of missing or incomplete data?

- A measurable portion of records contained missing region information, now categorized as “Unknown”. This suggests incomplete data capture during the signup process.

3. Are older users more or less likely to opt in to marketing?

- Users within the 25–40 age group show a higher marketing opt-in rate compared to older segments. Younger users appear more receptive to marketing communication.

4. Which plan is most commonly selected, and by which age group?

- The Basic plan is the most selected subscription tier overall. It is particularly prevalent among users aged 25–40. Premium plan selection increases slightly among older users.

5. (Optional) Which plan's users are most likely to contact support?

- A total of 27 customers contacted support within two weeks of signing up.

Pro plan users contacted support the most (12), followed by Basic (9) and Premium (6).

Regionally, Pro users in the North show the highest early support activity. Basic support requests are more evenly spread, with slightly higher numbers in the South and West. Premium support activity is mainly concentrated in the West.

This suggests that Pro users may require more assistance during the early stages of their subscription, particularly in certain regions.

5. Recommendations

1. Optimize High-Performing Channels:

Given Google's strong contribution to sign-ups, continued investment and optimization of search campaigns is recommended.

2. Strengthen Data Collection Controls:

Region and marketing fields should be validated at the point of entry. Dropdown menus and mandatory fields would reduce incomplete records.

3. Target Core Age Segment:

Marketing campaigns and onboarding messaging should align with the 25–40 age segment, which represents the largest share of customers.Upsell strategies may be refined for older segments showing interest in higher-tier plans.

6. Data Issues or Risks:

Several data quality risks were identified during the analysis.

Examples include:

• Invalid date formats in the signup_date column

• Text values appearing in numeric fields such as age

• Inconsistent category labels in plan_selected and gender fields

These issues increase data preparation time and may lead to inaccurate reporting if not addressed.

To improve data quality, the company should implement validation controls at the data entry stage. This could include mandatory structured fields, dropdown selections for categorical variables, and automated data quality checks within the data pipeline.